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The Spectrum and Structure of Singlet CH$_2$

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The spectrum and structure of singlet CH_2

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[Plates 20 to 22]

The red absorption bands of CH_2, previously shown to correspond to a transition between singlet states, have been analysed in detail. In the lower electronic state $\tilde{a}\,{}^1A_1$ the molecule is a strongly asymmetric top with rotational constants $A_{000} = 20{\cdot}14$, $B_{000} = 11{\cdot}16$ and $C_{000} = 7{\cdot}06$ cm^{-1}. From these constants a CH distance of 1·11 Å and an HCH angle of 102·4° is obtained. In the upper state $b\,{}^1B_1$ the molecule is nearly linear, having an HCH angle of about 140° and a CH distance of 1·05 Å. Most of the observed absorption bands correspond to excitation of the bending vibration ν_2' in the upper state ($v_2' = 6, 8, \ldots, 18$). In some of them in addition the symmetric stretching vibration ν_1 is singly excited. The levels 1, v_2–4, 0 cause strong perturbations in the levels 0, v_2, 0 for $v_2 = 8, 10, 12, 14$. In the lower state only one excited vibrational level 0, 1, 0 has been established which yields $\nu_2'' = 1352{\cdot}6$ cm^{-1}.

Fragments of some other singlet absorptions in the near ultraviolet and in the vacuum ultraviolet have been observed and are briefly described. The upper state of the near ultraviolet system is probably the predicted ${}^1A_1$ state corresponding to ${}^1\Sigma^+$ in the linear conformation.

1. Introduction

It was shown in the 1960 Bakerian lecture (Herzberg 1961) that there are two low lying states of the CH_2 radical, a singlet state (${}^1A_1$) in which the molecule is bent and a triplet state (${}^3\Sigma_g^-$) in which it is linear. The latter is in all probability the lower of the two, i.e. it is the real ground state of the molecule. In many photochemical reactions, however, because of spin conservation, CH_2 is first formed in a singlet state from which it is brought relatively slowly into the triplet ground state by collisions with other molecules. This transformation takes about 20 μs. Under our experimental conditions, this is the average life of CH_2 in the lowest singlet state, or, for short, of *singlet* CH_2.

The first strong absorption of triplet CH_2 lies in the vacuum ultraviolet at 1415 Å; but singlet CH_2 absorbs in the red and yellow part of the spectrum, giving rise to a large number of bands each consisting of many widely spaced fine lines. In the first work on this spectrum, in which only the three most obvious bands were analysed in a preliminary fashion (Herzberg 1961), it was shown that in the lower state of the transition the molecule is bent with an angle of about 103° similar to that in the ground states of H_2O and NH_2, while in the upper state it is linear or nearly linear.

In view of the interest attached to the CH_2 radical, it appeared desirable to attempt a more complete analysis of the red bands in order to obtain precise data about the structure of the molecule in the two states involved and to understand better the nature of the two electronic states. The present paper gives the results of such an analysis. At the same time, some studies of other singlet transitions of CH_2 will be described.

2. Experimental

The absorption spectrum of singlet CH_2 was obtained, just as that of triplet CH_2, by the flash photolysis of diazomethane (CH_2N_2) in the presence of a large excess of inert gas, usually N_2. The diazomethane was freshly prepared for each experiment by the action of aqueous KOH on *N*-nitrosomethyl urea dissolved in di-*n*-amyl ether. Several experiments were carried out with deuterated diazomethane obtained from deuterated *N*-nitroso-methyl urea (supplied by Merck, Sharp and Dohme of Canada, Ltd) and KOD. A few samples of diazomethane containing 60 % ^{13}C were prepared by Dr L. C. Leitch and enabled us to obtain spectra of $^{13}CH_2$.

In order to increase the intensity of the red CH_2 bands a White mirror system was used in an absorption tube of 2 m length making it possible to obtain an absorbing path length of up to 120 m. Photolysis in the quartz absorption tube was produced by two 1 m quartz flash lamps mounted end to end and connected in series. They were filled with xenon at 10 mm pressure. A condenser battery of 100 μF charged to 9000 V (i.e. about 4000 J) was discharged through these lamps. The duration of this flash was about 40 μs. The continuous background for the absorption was supplied by a smaller flash (about 225 J) through a quartz capillary filled with argon at 70 mmHg pressure and viewed sideways. The duration of the source flash was 15 μs. The maximum intensity of singlet CH_2 absorption was obtained when the source flash was fired somewhat before the photolysis flash attained its peak. As mentioned in the earlier paper the optimum mixing ratio of N_2 and CH_2N_2 for obtaining the red CH_2 bands was 100 : 1, quite different from the optimum ratio for the ultraviolet (triplet) spectrum of CH_2 for which it is 500 : 1 or even more.

In order to improve the intensity of the singlet CH_2 absorption still further, an attempt was made to shorten the durations of both photolysis and source flashes since it appeared that in this way, because of the shortness of the lifetime of singlet CH_2, the instantaneous concentration (which determines the absorption intensity if the source flash is sufficiently short) could be raised. In order to accomplish this shortening the self-induction in both circuits was drastically reduced (*a*) by using special low inductance capacitors, (*b*) by using flat copper conductors separated only by two sheets of Milar, and (*c*) by putting the return lead of the flash lamp as close as possible to the lamps. With these modifications, flash durations of 15 and 5 μs were obtained for photolysis and source flash respectively. For reasons not entirely understood the improvement of the singlet CH_2 absorption obtained by using these short flashes was disappointingly small, and most of the plates measured for the present analysis were those obtained with the old apparatus.

The spectrum of singlet CH_2 in the region 9200 to 4900 Å was photographed in the first and second orders of a 6·5 m concave grating spectrograph. The slit width used depended on the plate sensitivity in the region under study; it was usually 80 μm or less corresponding to a resolving power of 70 000 or better. An extension of the spectrum beyond 9200 Å where many further bands are expected was not possible because of the low sensitivity of *I–Z* plates which would have required an impractical number of flashes for one exposure. Fe lines from a hollow cathode discharge tube served as wavelength standards. The relative accuracy of the wavelengths of

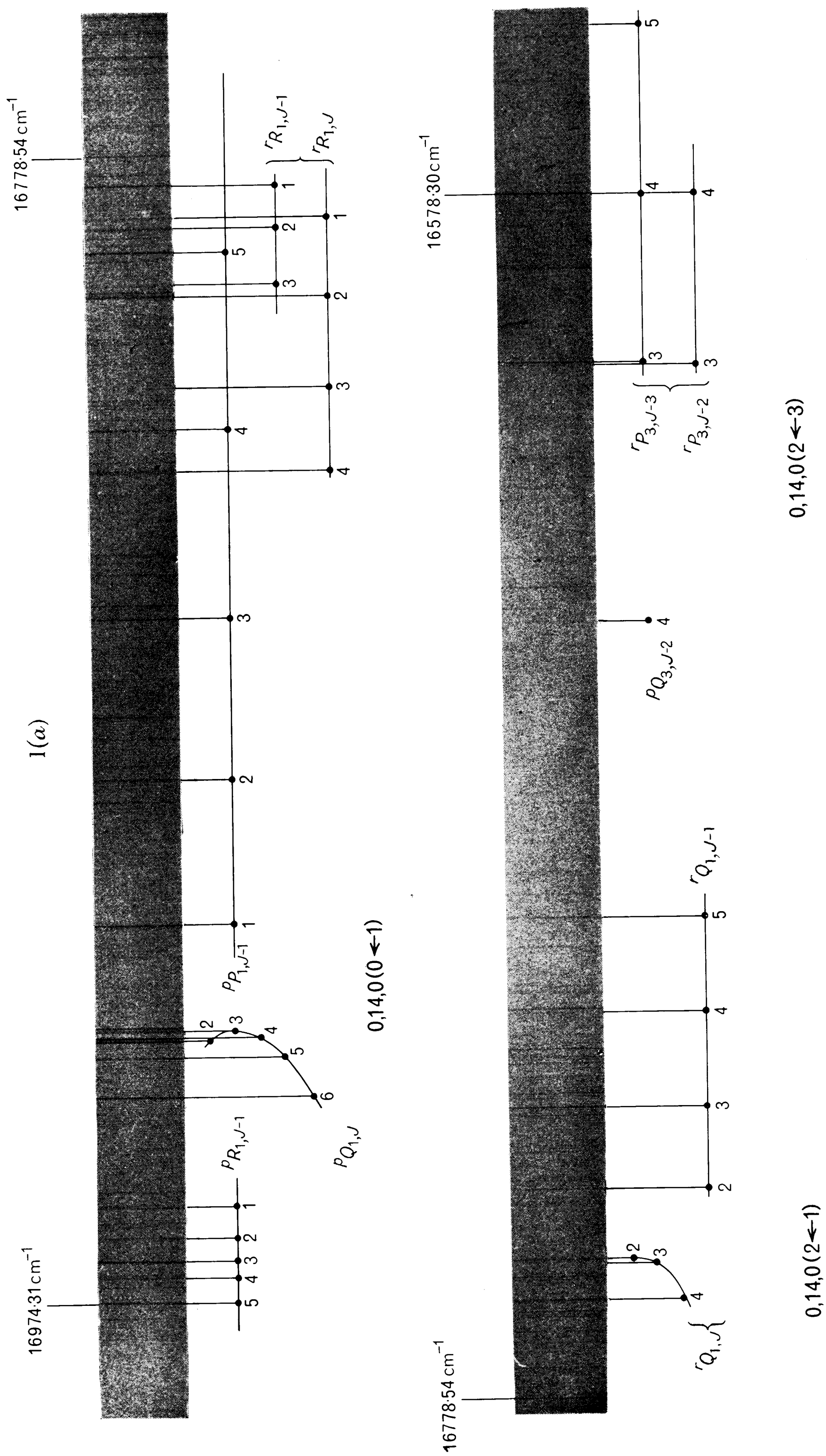

FIGURE 1 (*a*). Absorption band of CH_2 near 5900 Å. Assignments to the $\Sigma-\Pi$, $\Delta-\Pi$ and $\Delta-\Phi$ subbands are indicated.

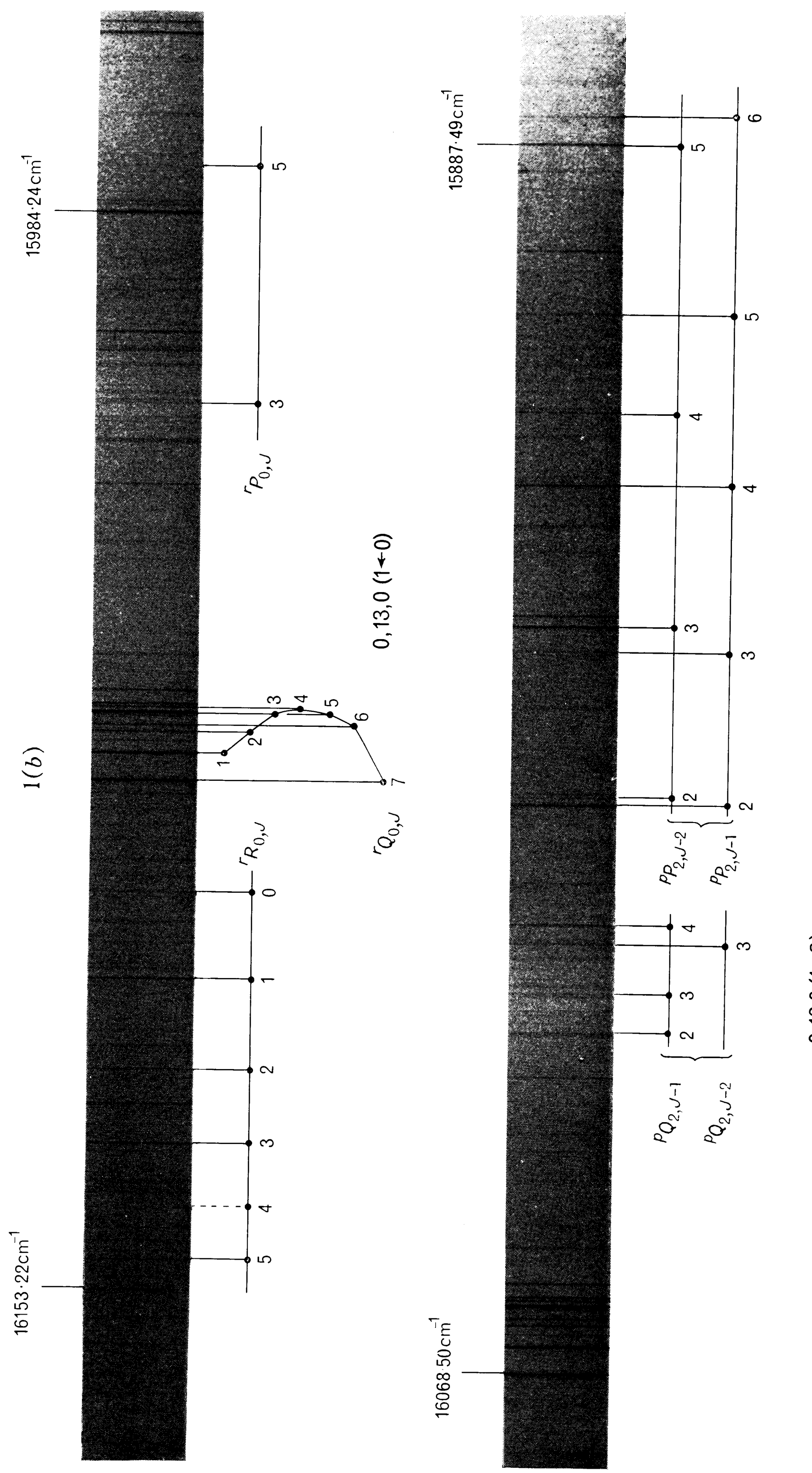

FIGURE 1 (*b*). Absorption band of CH_2 near 6220 Å. Assignments to the Π–Σ and Π–Δ subbands are indicated.

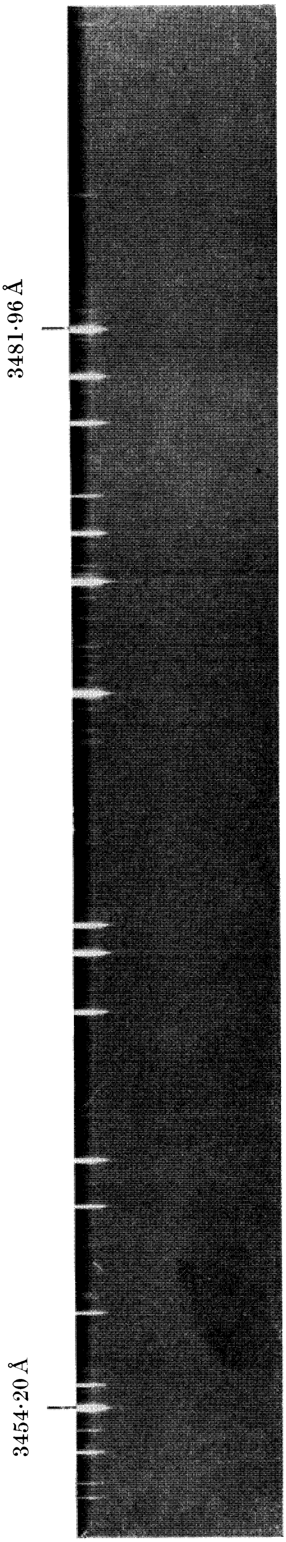

FIGURE 2. Absorption band of CH_2 near 3470 Å. This is the strongest of the near ultraviolet bands.

unblended lines measured on one plate is estimated to be $\pm 0{\cdot}02\,cm^{-1}$, but the absolute accuracy is substantially less (perhaps $\pm 0{\cdot}1\,cm^{-1}$), since during some of the long exposures shifts caused by temperature and barometric pressure changes occurred.

An extensive search was made for other transitions of CH_2 in the region 5000 to 1300Å. For the search below 2000Å, the flash photolysis apparatus was coupled to a 3 m vacuum spectrograph in the manner described in the earlier paper. Above 1500Å, three traversals through a 2 m tube were possible. Only fragments of other singlet systems were found (see § 3).

3. Observed spectra

The red absorption spectrum of singlet CH_2 is a many line spectrum consisting of hundreds of sharp lines which are only indistinctly grouped in bands. This appearance is very similar to that of the red NH_2 bands analysed by Dressler & Ramsay (1959). In figure 1 (*a*) and (*b*), plates 20 and 21, two sections of the CH_2 spectrum are shown. With the increased path length, compared to the earlier work, the spectrum was extended to 4900Å on the short wavelength side.

The isotope investigations with ^{13}C and deuterium substituted diazomethane described in the earlier paper have established that CH_2 is the molecule responsible for this spectrum. At that time also three bands were measured in each of which three branches, P, Q and R, were fairly easily recognized. With the new plates these branches were considerably extended, and further similar bands were found at shorter and longer wavelengths. Several of these form a progression with a spacing of about $1600\,cm^{-1}$. The branches of all these bands are listed in table 1 (*a*).

Halfway between the bands just discussed, equally strong bands of a different type appear: they consist each of two subbands both having P, Q and R branches, but with the P branch weak in the one and the R branch weak in the other. In addition, in one subband the branches start with $J = 0$, in the other with $J = 2$, while the branches of the first type of band start with $J = 1$. In other words, bands of the first type are similar to Σ–Π bands of linear molecules, while bands of the second type are similar to Π–Σ and Π–Δ bands. The branches of the bands belonging to the second type are listed in table 1 (*b*). Finally, there are weaker bands close to the bands of the first and second type in which an increasing number of lines are missing at the beginning of the branches and which are similar to Δ–Π, Δ–Φ, Γ–Φ, Γ–H and Φ–Δ, Φ–Γ bands of linear molecules. The lines of these bands are listed in table 1 (*c*), (*d*) and (*e*). In addition to the main progression, there are a number of weaker bands of similar structure. The lines of these bands are included in table 1. Their vibrational assignment will be discussed in § 4.

The spectra obtained with deuterated diazomethane were very much weaker than those obtained with ordinary diazomethane. They were not considered sufficient for a detailed analysis and were therefore not measured. No new spectra of $^{13}CH_2$ were taken. The spectra taken and illustrated in the earlier work, while entirely adequate to establish the presence of carbon in the molecule, are not suitable for a detailed analysis. However, the isotope shifts of the stronger features were obtained from enlarged prints of these spectrograms.

Lines which remain unassigned in the red region of the spectrum have been listed in table 2. They may either be due to hot bands (other than those arising from 0, 1, 0; see § 5) or correspond to unrecognized perturbed levels of the excited state.

In addition to the red absorption bands, there are a few bands of similar structure in the near ultraviolet (3600 to 3300Å). Even under the best conditions, these bands were rather weak. Figure 2, plate 22, shows the strongest of these. Branches are not readily recognized. The observed lines are listed in table 3.

Table 1. Vacuum wavenumbers of the rotational lines of the $\tilde{b}\ {}^1B_1\Delta_g$–$\tilde{a}\ {}^1A_1$ system of CH_2

(*a*) *Σ vibronic levels* (0–1 *subbands*)

	transition	1, 4, 0–0, 0, 0	0, 6, 0–0, 0, 0	1, 6, 0–0, 0, 0	0, 8, 0–0, 0, 0	1, 8, 0–0, 0, 0	0, 10, 0–0, 0, 0
R	2_{02}–1_{10}	—	—	13 848·55	12 234·76	—	13 693·31
	3_{03}–2_{11}	—	—	—	240·52	—	700·02
	4_{04}–3_{12}	12 451·19	—	854·28	242·47	15 136·50	704·86
	5_{05}–4_{13}	—	—	—	241·95*	—	—
	6_{06}–5_{14}	—	—	848·56	241·95*	—	—
Q	1_{01}–1_{11}	—	10 811·4	—	12 207·87	—	13 666·34*
	2_{02}–2_{12}	12 414·16	812·9	13 820·28	206·55*	15 100·08	665·03
	3_{03}–3_{13}	—	—	818·78	205·02	—	664·46
	4_{04}–4_{14}	412·90*	—	815·84	204·06*	097·97	666·34*
	5_{05}–5_{15}	—	—	814·16	204·06*	—	—
	6_{06}–6_{16}	—	—	813·29	206·55*	096·24	—
P	0_{00}–1_{10}	—	—	13 802·78	12 188·27	—	13 646·61
	1_{01}–2_{11}	—	10 767·6	779·96	163·22	—	621·48
	2_{02}–3_{12}	12 341·74	—	747·88	134·07	15 027·78	592·69
	3_{03}–4_{13}	—	—	—	102·41	—	561·80
	4_{04}–5_{14}	278·71	—	681·94	070·05	14 964·05	532·43
	5_{05}–6_{15}	—	—	—	039·15	—	—
	6_{06}–7_{16}	—	—	620·40	010·72	913·20	—

	transition	1, 10, 0–0, 0, 0	0, 12, 0–0, 0, 0	0, 14, 0–0, 0, 0	0, 14, 0–0, 1, 0	0, 16, 0–0, 0, 0	0, 16, 0–0, 1, 0
R	2_{02}–1_{10}	—	15 334·07	16 957·69	—	18 625·86	—
	3_{03}–2_{11}	—	340·62	963·20	—	632·84	—
	4_{04}–3_{12}	—	344·22	967·25	15 612·30	636·21	17 281·18
	5_{05}–4_{13}	—	347·21	970·00	—	—	—
	6_{06}–5_{14}	—	352·96	974·31	—	637·97	281·38
Q	1_{01}–1_{11}	—	15 306·96	16 929·33*	—	18 598·40	—
	2_{02}–2_{12}	—	305·76*	929·33*	15 575·40	597·65*	17 243·53*
	3_{03}–3_{13}	—	305·05	927·65	574·11	597·26	243·53*
	4_{04}–4_{14}	16 473·44	305·76*	928·79	575·80	597·65*	244·58
	5_{05}–5_{15}	—	309·36	932·19	—	599·51	—
	6_{06}–6_{16}	—	317·62	939·01	587·06	602·78	250·66
P	0_{00}–1_{10}	—	15 287·32	16 909·39	—	18 578·45	17 223·93
	1_{01}–2_{11}	—	262·18	884·47*	—	553·67	—
	2_{02}–3_{12}	—	233·39	856·96	15 501·93	525·18	170·10
	3_{03}–4_{13}	—	202·44	825·05	—	494·70	—
	4_{04}–5_{14}	16 339·44	171·79	794·80	438·12	463·74	106·98
	5_{05}–6_{15}	—	144·40	767·31	—	—	—
	6_{06}–7_{16}	—	—	743·52	—	407·15	—

* Denotes a line used more than once in the analysis.

Table 1 (*cont.*)

(*b*) *Π vibronic levels* (1–0 *subbands*)

	transition	(*a*)– 0, 0, 0	0, 9, 0– 0, 0, 0	0, 11, 0– 0, 0, 0	(*b*)– 0, 0, 0	0, 13, 0– 0, 0, 0	0, 15, 0– 0, 0, 0	0, 15, 0– 0, 1, 0
R	1_{10}–0_{00}	12 647·54	12 987·40	14 407·38	—	16 082·91	17 739·04	—
	2_{11}–1_{01}	669·46	13 001·42	421·13	14 607·90	098·05	757·63	16 405·25
	3_{12}–2_{02}	—	010·48	—	—	111·63	773·85	—
	4_{13}–3_{03}	692·19	019·12	447·34	642·48	122·86	794·29	—
	5_{14}–4_{04}	—	—	—	—	—	—	—
	6_{15}–5_{05}	—	041·87	—	—	138·90	843·16	—
Q	1_{11}–1_{01}	12 637·10	12 970·09	14 388·83	14 578·00*	16 064·40	17 724·36	16 371·94
	2_{12}–2_{02}	640·82	972·99	384·90	—	061·20	—	—
	3_{13}–3_{03}	648·08	973·71	381·56	587·94	058·29*	704·84	352·41
	4_{14}–4_{04}	—	977·25	380·39	—	057·53	—	—
	5_{15}–5_{05}	—	981·23	379·15	—	058·29*	700·68	348·69
	6_{16}–6_{06}	—	—	—	—	060·23	—	—
	7_{17}–7_{07}	—	—	—	—	068·50	—	—
P	2_{11}–3_{03}	—	12 915·39	—	—	16 011·97*	—	—
	(1–2 *subbands*)							
Q	2_{11}–2_{21}	12 589·18	12 921·13	14 340·78	14 527·76	16 017·80	17 677·37	—
	2_{12}–2_{20}	—	—	—	—	—	—	—
	3_{12}–3_{22}	—	910·82	—	—	16 004·16	17 650·64	—
	3_{13}–3_{21}	—	—	—	—	—	—	—
	4_{13}–4_{23}	—	898·05	326·28	—	001·83	—	—
P	1_{10}–2_{20}	—	12 887·21	14 307·72	—	15 983·15	17 639·56*	—
	1_{11}–2_{21}	12 556·87	889·83	308·61	14 497·85	984·24	644·09	16 284·99
	2_{11}–3_{21}	529·20	861·14	280·84	467·61	957·89	617·34	258·23
	2_{12}–3_{22}	541·21	873·29	285·26	—	961·59	—	—
	3_{12}–4_{22}	—	—	—	—	926·58	588·72	—
	3_{13}–4_{23}	527·00	852·60	260·54	466·67	937·24	583·80	—
	4_{13}–5_{23}	—	783·78	211·92	407·07	887·49*	558·82	—
	4_{14}–5_{24}	—	—	—	—	912·17	—	—
	5_{14}–6_{24}	—	—	—	—	—	—	—
	5_{15}–6_{25}	—	809·68	208·36	—	887·49*	529·72	—
	6_{15}–7_{25}	—	692·19	—	—	789·25	—	—
	6_{16}–7_{26}	—	—	—	—	863·88	—	—
	7_{16}–8_{26}	—	—	—	—	—	—	—
	7_{17}–8_{27}	—	—	—	—	841·23	—	—

Table 1 (*cont.*)

(*c*) *Δ vibronic levels* (2–1 *subbands*)

	transition	0, 8, 0–0, 0, 0	0, 10, 0–0, 0, 0	(*c*)–0, 0, 0	0, 12, 0–0, 0, 0	(*d*)–0, 0, 0	0, 14, 0–0, 0, 0	0, 16, 0–0, 0, 0
R	2_{20}–1_{10}	11 740·27	13 316·04	13 663·10	15 262·94	15 085·81	16 783·57	18 418·16
	2_{21}–1_{11}	744·29	—	—	—	—	788·35	—
	3_{21}–2_{11}	749·78	—	—	—	—	790·88*	—
	3_{22}–2_{12}	761·63	338·85	687·86	285·42	109·19	802·24	440·42
	4_{22}–3_{12}	756·73	—	—	—	—	800·38	437·66
	4_{23}–3_{13}	779·78	—	—	—	—	816·85	
	5_{23}–4_{13}	763·85	—	—	—	—	—	—
	5_{24}–4_{14}	799·56	—	—	—	—	831·98	—
	6_{24}–5_{14}	769·79	—	—	—	—	—	—
Q	2_{20}–2_{12}	11 712·09	13 287·73	13 635·01	15 234·68	—	755·27*	18 389·96
	2_{21}–2_{11}	699·51	—	—	—	—	743·52	—
	3_{21}–3_{13}	714·12	—	—	—	—	755·27*	—
	3_{22}–3_{12}	689·22	266·42	615·46	212·94	15 036·74	729·77	367·98
	4_{22}–4_{14}	718·27	—	—	—	—	761·97	399·27
	4_{23}–4_{13}	677·27	—	—	—	—	713·92	—
	5_{23}–5_{15}	—	—	—	—	—	—	—
	5_{24}–5_{14}	655·59	—	—	—	—	697·98	—
	(2–3 *subbands*)							
Q	3_{22}–3_{30}	11 612·43	—	13 538·65	15 136·15	14 959·99	16 652·93	—
	4_{22}–4_{32}	605·23	—	—	—	—	648·85	—
P	2_{20}–3_{30} 2_{21}–3_{31}	11 562·82	13 138·55	13 485·71	15085·43	14 908·47	16 606·10 607·07	18 240·72
	3_{21}–4_{31} 3_{22}–4_{32}	537·30 537·73	114·91	463·99	061·39	885·39	578·30	216·44
	4_{22}–5_{32}	507·07	—	—	—	—	550·71	188·06
	4_{23}–5_{33}	510·56	—	—	—	—	547·25	—
	5_{23}–6_{33}	473·16	—	—	—	—	—	—
	5_{24}–6_{34}	481·90	—	—	—	—	514·30	—
	6_{24}–7_{34}	450·34	—	—	—	—	—	—

(*d*) *Φ vibronic levels* (3–2 *subbands*)

	transition	0, 9, 0–0, 0, 0	0, 11, 0–0, 0, 0	(*e*)–0, 0, 0	0, 13, 0–0, 0, 0	0, 15, 0–0, 0, 0
R	3_{30}–2_{20}	12 223·63	—	—	—	—
	3_{31}–2_{21}	224·45	13 781·22	14 262·67	15 849·29	17 417·92
	4_{31}–3_{21}	230·13	787·70	275·48	861·16	427·87
	4_{32}–3_{22}	236·34	793·47	—	—	—
	5_{32}–4_{22}	—	—	—	—	—
	5_{33}–4_{23}	247·19	804·45	—	883·47	450·97
Q	3_{31}–3_{21}	12 164·45	—	14 202·39	15 789·25	17 357·97
	4_{31}–4_{23}	161·81	—	—	—	—
	4_{32}–4_{22}	150·95	—	—	—	—
	5_{33}–5_{23}	132·91	13 690·19	—	769·11	336·73
	(3–4 *subbands*)					
P	3_{30}–4_{40} 3_{31}–4_{41}	11 966·93	13 523·79	14 005·28	15 591·82	17 160·38
	4_{31}–5_{41}	938·74	496·40	13 984·03	569·79	136·65
	4_{32}–5_{42}	940·09	497·26	—	—	—
	5_{32}–6_{42}	—	—	—	—	—
	5_{33}–6_{43}	908·58	465·98	—	544·95	112·53

Table 1 (*cont.*)

(*e*) Γ *vibronic levels* (4–3 *subbands*)

	transition	0, 8, 0– 0, 0, 0	0, 10, 0– 0, 0, 0	0, 12, 0– 0, 0, 0	0, 14, 0– 0, 0, 0	0, 16, 0– 0, 0, 0
R	4_{40}–3_{30}	11846·92	12660·76	14305·72	16338·36	17945·27
	4_{41}–3_{31}	—	660·95	—	338·64	—
	5_{41}–4_{31}	—	665·51	—	—	—
	5_{42}–4_{32}	—	666·62	312·18	354·31	—
	6_{42}–5_{32}	—	669·46	—	—	—
Q	4_{40}–4_{32}	11772·34	12586·04	—	16263·64	17870·58
	4_{41}–4_{31}	—	581·22	—	—	—
	5_{42}–5_{32}	—	568·44	—	256·21	—
			(4–5 *subbands*)			
P	4_{40}–5_{50}, 4_{41}–5_{51}	11513·13	12326·94	13971·90	16004·64	17611·45
	5_{41}–6_{51}	—	295·38	—	—	—
	5_{42}–6_{52}	—	294·91	940·45	15982·54	—
	6_{42}–7_{52}	—	264·96	—	—	—

Finally, in the vacuum ultraviolet near the triplet absorption a few bands were found which were strong when the conditions favoured singlet CH_2. The wavelengths of the most prominent features are 1400·9 (*s*), 1403·6 (*w*), 1403·8 (*w*), 1404·4 (*m*) and 1418·0 (*s*). One of these features (at 1418·0 Å) shows a rotational structure very similar to that of the main triplet band at 1415·6 Å except that the lines are much less diffuse. It seems probable that this band is a hot band of triplet CH_2. In view of this interpretation it remains somewhat uncertain whether the other features near 1400 Å are due to singlet CH_2.

4. Rotational analysis

In the earlier work, from a preliminary analysis of the three strongest bands, it was concluded that the CH_2 molecule is linear or nearly linear in the upper state of the red bands but strongly bent in the lower state. These conclusions have been fully confirmed by the detailed analysis of the new spectra. The structure of the bands of the first type (similar to Σ–Π bands) is readily understood on this basis. Although in the lower state the molecule is an asymmetric top, the quantum number K_a of the angular momentum about the a axis is still quite well defined and its value ($K_a = 0, 1, 2, \ldots$) indicates whether a lower state sublevel behaves like a Σ, Π, Δ, … vibronic level of a linear molecule. The large combination defect in the bands of the first type is caused by the large K type (asymmetry) doubling in the lower state. The magnitude of this doubling (for $K_a = 1$) is in a first approximation given by

$$\Delta\nu_{cd} = \tfrac{1}{2}(B-C)\,J(J+1). \tag{1}$$

Since, in addition, the average of the two components gives $\bar{B} = \frac{1}{2}(B+C)$, we obtain the rotational constants B'' and C'' separately. From them, on the assumption of zero inertial defect, a preliminary value of A'' can be determined. From the three rotational constants, preliminary values of the rotational levels of the lower state

TABLE 2. UNASSIGNED LINES IN THE SPECTRUM OF CH_2

$\nu_{vac.}$	I	$\nu_{vac.}$	I	$\nu_{vac.}$	I
18753·40	2	16854·63	5	14961·39	3
749·45	2	801·58	2	847·10	4
650·90	2	778·54	3	821·61	4
554·52	1	763·91	3	817·42	3
501·43	1	713·17	3	816·34	3
469·32	2	590·02	2	813·57	3
466·99	2	379·52	1	812·09	6
459·46	1	364·62	2	804·38	2
457·02	1	363·47	2	802·02	2
445·10	3	324·70	5	578·00	2
407·15	2	317·70	4	426·64	4
389·12	1	191·60	4	417·80	4
367·00	1	153·22	3	404·21	3
360·97	1	144·80	2	382·81	5
211·72	1	116·82	3	349·32	3
042·35	2	096·61	3	334·64	4
041·40	2	089·56	3	208·36	3
036·42	3	077·45	2	077·34	3
035·18	3·	057·02	5	075·81	3
023·32	3	055·01	5	067·93	2
022·46	2	024·40	4	13744·17	2
17987·42	2	15956·30	5	726·14	4
959·31	4	943·09	3	652·94	6
858·61	2	902·96	4	642·54	6
855·61	1	838·26	5	610·24	4
779·76	3	799·03	4	307·59	2
740·06	3	794·14	4	138·55	4
732·75	3	781·56	2	12959·40	6
723·61	2	780·27	2	954·75	2
722·58	2	715·96	3	953·55	2
707·24	4	707·58	3	949·56	3
704·44	3	608·48	5	882·84	4
703·77	3	483·05	4	870·44	4
700·16	2	375·88	3	869·68	4
699·46	2	354·42	3	857·22	4
698·84	2	350·82	3	839·53	6
694·40	2	341·12	2	754·11	4
660·34	3	328·34	6	751·23	4
658·62	3	327·86	4	711·81	6
642·80	2	290·79	3	692·45	4
633·10	1	266·93	6	692·19	4
481·71	2	255·12	4	674·87	6
466·32	2	207·88	3	202·31	5
427·87	3	094·47	2	11931·54	6
363·63	2	087·16	4	734·49	4
361·02	4	075·16	6	721·31	3
109·23	3	065·62	5	720·48	3
073·38	2	024·06	4	703·81	4
16927·06	5	020·76	4	504·16	6

were derived. Thus it was possible to predict the approximate structure of the Π–Σ and Π–Δ type bands and the combination differences $\Delta F(J, K_a)$ that must occur in them. As an example, table 4 shows the differences

$$\Delta_{2,2}F(J, K_a) = F''(J+1, 2) - F''(J-1, 0) \qquad (2)$$

predicted in this way and used to find the Π–Σ and Π–Δ type subbands. For comparison, in the last column of table 4 the observed combination differences obtained from the assigned lines are given.

TABLE 3. VACUUM WAVENUMBERS AND RELATIVE INTENSITIES OF LINES PROBABLY DUE TO SINGLET CH_2 BETWEEN 3330 AND 3625 Å

$\nu_{vac.}$	I	$\nu_{vac.}$	I	$\nu_{vac.}$	I
30 035·37	1	782·70	2	747·50	1
032·66	1	779·08	2	715·42	3
030·77	1	728·43	1	684·42	1
026·16	2	28 873·43	2	27 789·94	1
024·40	2	868·10	2	776·73	2
008·15	2	858·82	1	765·68	3
004·63	2	855·30	1	750·31	3
29 995·80	1	850·28	2	747·49	3
992·69	1	847·39	1	738·33	1
990·42	2	845·96	3	721·80	4
987·93	1	843·65	1	719·22	2
984·18	2	795·95	3	707·11	2
976·46	1	787·62	1	699·83	1
926·00	1	782·07	2	688·52	3
915·21	1	778·78	2	685·77	1
858·87	1	774·78	2	683·49	2
855·07	2	765·81	2	639·50	2
830·47	2	763·76	1	611·28	2
818·41	0	763·06	3	596·47	3
816·36	0	757·63	2	586·84	2
787·71	1	752·53	2		

TABLE 4. GROUND STATE COMBINATION DIFFERENCES IN Π–Σ AND Π–Δ SUBBANDS

difference	estimate	found
2_{20}–0_{00}	98·1	99·66, 99·69, 99·66
2_{21}–1_{01}	78·7	80·25, 80·29, 80·28, 80·26, 80·35, 80·16, 80·26, 80·23, 80·22
3_{21}–1_{01}	138·7	140·16, 140·28, 140·26, 140·29, 140·29, 140·29
3_{22}–2_{02}	98·1	99·66, 99·66, 99·61, 99·70, 99·61, 99·64
4_{23}–3_{03}	119·6	121·03, 121·07, 121·06, 121·00, 121·05, 121·11, 121·08, 121·02
5_{23}–3_{03}	234·3	235·37, 235·34, 235·25, 235·42
5_{05}–3_{03}	143·7	145·60, 145·83
6_{25}–5_{05}	169·7	171·55, 170·80?
7_{25}–5_{05}	350·1	349·68, 349·61

In a similar way the other bands involving Δ, Φ, Γ upper vibronic states were analysed. At all stages of the analysis the assignments were checked by combination relations for the lower state: all Π–Σ, Π–Δ bands must give the same values (within the accuracy of the measurements) of the combination differences, and similarly for all other types of bands. No assignment was considered as proved until confirmed by agreement of at least one pair of combination differences. In addition, the combination differences had to be close to the values predicted from the provisional rotational constants but allowing for the effects of centrifugal distortion.

In all branches there is a strong intensity alternation on account of the presence of two identical nuclei. For $K'' = 0$ the strong lines are those with odd J. For a C_{2v} molecule in which the twofold axis of symmetry coincides with the b axis, such an alternation shows that the lower state is A_1 or A_2. The latter alternative can be excluded on the basis of the electron configuration (see §7).

The fact that only transitions with $\Delta K = l' - K''_a = \pm 1$ were observed shows that the transition moment is perpendicular to the a axis in the lower state. In addition, from the observation that the Q lines of the Σ–Π type subbands arise from the lower K type doubling component, it follows unambiguously (see Herzberg 1945) that the transition moment is perpendicular to the plane of the molecule (type C bands), that is, that the upper state of the transition is B_1 (on the assumption that the lower state is A_1, see §7).

All the lines of the branches of the red bands are single, and therefore we must conclude that the red bands represent a singlet transition, namely ${}^1B_1$–${}^1A_1$.

With the rotational levels of the lower state of the red bands established, we expected that it would be a simple matter to assign the lines of the near ultraviolet bands, since it was fairly certain from the characteristic appearance of the bands, and from the way in which the bands decayed with time, that they had the same lower state as the red bands. However, the analysis proved to be quite difficult because of the limited number of observed lines. Only a few combination differences were found which agree with those already known for the lowest singlet state of CH_2. Since the only expected singlet state in the energy range 2 to 5 eV is ${}^1A_1$ (see §7) we conclude that the upper state of the near ultraviolet bands is ${}^1A_1$, but the few observed lines are not sufficient to prove this conclusion unambiguously. Nor are they sufficient to determine the rotational constants of the upper state.

No rotational analysis of the vacuum ultraviolet singlet bands was possible.

5. Vibrational analysis

The wavenumbers of the origins of all the observed subbands are listed in table 5. In the main progression, eleven bands have been observed, of which the strongest subbands are alternately of type Σ–Π and Π–Σ. Subbands of types Δ–Π, Δ–Φ, ... and Π–Δ, Φ–Δ, ... have been observed for the stronger of the main bands. The alternation of even and odd l' in the main progression shows immediately that this progression must correspond to the excitation of successive quanta of the bending vibration ν_2 in the upper state and that the molecule must be effectively linear in this state. Indeed in a linear-bent transition the progression in ν_2 is expected to be the most prominent one. The magnitude of ν_2 obtained from the spacing of successive bands (about 750 cm^{-1}) is reasonable for an H—C—H bending vibration.

The vibrational numbering of the bands in the main progression is not obvious since the intensity maximum is far from the 0, 0, 0–0, 0, 0 band. However, the isotope shifts between ${}^{13}CH_2$ and ${}^{12}CH_2$ can aid in determining the numbering. In figure 3 these shifts are plotted against band position. For the correct numbering the nearly straight line should go through zero at the position of the 0, 0, 0–0, 0, 0 band, except for the effect of zero point energy. Unfortunately because of perturbations (see below) the slope of the line is not well defined and leaves a wide margin of uncertainty for the position of the 0, 0, 0–0, 0, 0 band. But according to elementary

Table 5. Constants (in cm^{-1}) of the $\tilde{b}\ {}^1B_1$ state of CH_2

(*a*) Σ *vibronic levels*

v_1, v_2, v_3	ν_0	B_v	$D_v \times 10^3$
0, 6, 0	$10823{\cdot}_2$	7·7	−1·8
0, 8, 0	12219·6	7·72	−12
0, 10, 0	13677·9	7·76	−6·5
0, 12, 0	15318·6	7·79	−2·3
0, 14, 0	16940·7	7·84	−6·1
0, 16, 0	18609·8	7·89	−1·6
0, 18, 0	20302	—	—
1, 4, 0	$12428{\cdot}_2$	7·59	−0·6
1, 6, 0	13834·1	7·63	−6
1, 8, 0	15113·8	7·64	−5
1, 10, 0	16490	—	—

(*b*) Π *vibronic levels*

v_1, v_2, v_3	ν_0	B	q
(*a*)	12638·2	8·8	−1·8
0, 9, 0	12972·3	$7{\cdot}9_0$	$-1{\cdot}1_4$
0, 11, 0	14391·3	$7{\cdot}9_5$	−0·14
(*b*) or 1, 7, 0?	14578	—	—
0, 13, 0	16066·5	8·0	*ca.* −0·04
0, 15, 0	17721·3	$8{\cdot}00_4$	$+1{\cdot}36_5$

(*c*) Δ *vibronic levels*

v_1, v_2, v_3	ν_0	B
0, 8, 0	$11721{\cdot}_3$	8·3
0, 10, 0	$13296{\cdot}_0$	8·5
(*e*) or 1, 6, 0?	$13641{\cdot}_3$	8·8
(*d*) or 1, 8, 0	$15065{\cdot}_2$	8·6
0, 12, 0	$15244{\cdot}_2$	$8{\cdot}4_5$
0, 14, 0	16764	8·4
0, 16, 0	18398	8·4

(*d*) Φ *vibronic levels*

v_1, v_2, v_3	ν_0	B
0, 9, 0	$12224{\cdot}_7$	$8{\cdot}_2$
0, 11, 0	$13779{\cdot}_0$	$8{\cdot}_4$
(*e*) or 1, 7, 0?	14261	—
0, 13, 0	15848	—
0, 15, 0	17416	—

(*e*) Γ *vibronic levels*

v_1, v_2, v_3	ν_0	B
0, 8, 0 or 1, 4, 0	11893	—
0, 10, 0	12707	$8{\cdot}_1$
0, 12, 0	14352	$8{\cdot}_1$
0, 14, 0	16384	—
0, 16, 0	17991	—

8-2

isotope theory (see Herzberg 1945, p. 230) and assuming CH_2 to be linear in the $\tilde{b}^1B_1$ state and neglecting anharmonicity, the slope of the line should be

$$1-\sqrt{\left[\frac{(m_{13}+2m_{\mathrm{H}})\,m_{12}}{(m_{12}+2m_{\mathrm{H}})\,m_{13}}\right]}=0{\cdot}0055_6, \tag{3}$$

where m_{12} and m_{13} are the masses of ^{12}C and ^{13}C. If the best straight line of this slope is drawn through the points of figure 3 one obtains for the position of the 0, 0, 0–0, 0, 0 band $7400 \pm 500\ \mathrm{cm^{-1}}$. This position of the system origin establishes the vibrational numbering adopted in the tables to within ± 2 units of v_2.

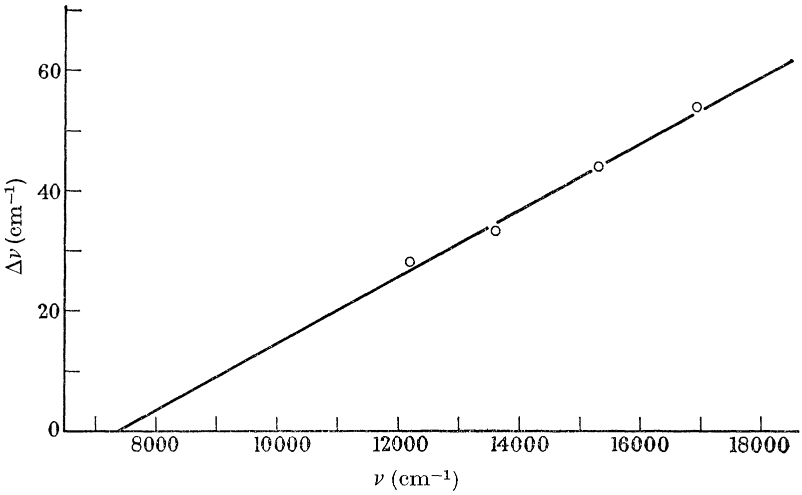

FIGURE 3. Isotope shifts between $^{12}CH_2$ and $^{13}CH_2$. The average shift of individual rotational lines in the Σ subbands has been plotted against the energy.

The spacing of the bands in the main progression is not quite regular. Such an irregularity can arise on account of Fermi resonance with other vibrational levels of the excited state. Indeed, four other Σ–Π type subbands have been found (one of them only a fragment) which are readily interpreted as the progression

$$1, v_2\text{–}4, 0 \leftarrow 0, 0, 0 \quad \text{with} \quad v_2 = 8, 10, 12, 14. \tag{4}$$

The levels $1, v_2$–$4, 0$ are close to $0, v_2, 0$ because $\nu_1 \simeq 4\nu_2$. Actually, because of the strong anharmonicity, the two unperturbed progressions $0, v_2, 0$ and $1, v_2$–$4, 0$ intersect and it is only in the neighbourhood of the point of intersection that the $1, v_2$–$4, 0$ progression has appreciable intensity. However, not all deviations from a smooth ΔG curve are accounted for in this way. It is clear that other resonances are possible and may be responsible for these deviations.

For $l' > 0$, unlike $l' = 0$, perturbations may also be produced by the higher vibrational levels of the ground state. The extra Π–Σ type subband at $12\,638\ \mathrm{cm^{-1}}$ may be caused by such a perturbation since its B' value is substantially larger than that of the other Π–Σ subbands. Other examples may be the subbands at 14 578, 13 641, 15 065, 14 261 and 11 893.

Because of the perturbations and because of the presence of a potential barrier (see below), it is not surprising that the vibrational levels cannot be represented very well by a standard formula of the type

$$G_0(v_2) = \omega_2^0 v_2 + x_{22}^0 v_2 + g_{22} l^2 + \ldots. \tag{5}$$

In figure 4 the deviations of the observed Σ, Π, Δ, Φ and Γ sublevels from the formula

$$T_0(v_2) = 7090 + 560 v_2 + 10 v_2^2 - 52 l^2 + 1 \cdot {}_0 l^4 \tag{6}$$

are plotted as ○, ×, △, Φ, and □ respectively. The 1, v_2–4, 0 sublevels are plotted at $v_2 + 4$ in order to make them comparable with the 0, v_2, 0 levels. It is seen that with the chosen formula the $\Pi, \Delta, \ldots$ levels for $v_2 \geqslant 12$ fall fairly closely on the same curve as the Σ levels, that is, the term $-52l^2 + 1 \cdot {}_0 l^4$ gives a fair representation of the dependence on l. However, for $v_2 < 12$ there are large deviations.

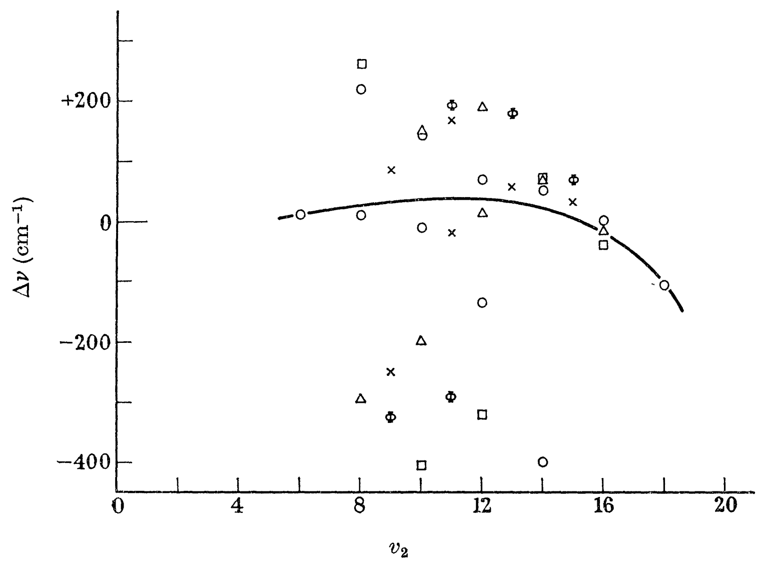

FIGURE 4. Deviations of observed subband origins from those calculated by the formula $T_0(v_2, l) = 7090 + 560 v_2 + 10 v_2^2 - 52 l^2 + 1{\cdot}0 l^4$. The Σ, Π, Δ, Φ and Γ sublevels have been plotted as ○, ×, △, Φ and □ respectively.

Since the perturbing Σ levels 1, v_2–4, 0 have been observed, we can, using well known formulae, readily determine the unperturbed positions of the Σ levels 0, v_2, 0. Figure 5, in which $\Delta_2 G$ is plotted as a function of v_2 gives the result of this deperturbation. The $\Delta_2 G$ values vary now in a fairly smooth manner. According to Dixon (1964), the fact that this curve tends towards a minimum for small v_2 suggests strongly that there is a maximum in the potential function of the bending vibration, or in other words, that in the equilibrium position of the $\tilde{b}$ state the molecule is slightly bent. We have tried to fit the observed (deperturbed) levels both to a potential function with a Gaussian maximum (following Dixon)

$$V = \tfrac{1}{2} k r^2 + \alpha \, \mathrm{e}^{-\beta r^2} \tag{7}$$

and one with a Lorentzian maximum (following Thorson & Nakagawa 1960)

$$V = \tfrac{1}{2} k r^2 + K_B / (C^2 + r^2), \tag{8}$$

where r is the coordinate of the bending vibration.

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Unfortunately the data are insufficient to determine uniquely the three parameters of either function. However, the energy levels can be represented very well, for example, by a potential with a Lorentzian maximum and with the following constants:

$$\omega_2 = 1039{\cdot}_7\,\mathrm{cm^{-1}}, \quad K_B = 372\,712\,\mathrm{cm^{-1}} \quad \text{and} \quad C^2 = 20. \tag{9}$$

The vibrational intervals obtained from this potential are also shown in figure 5 as a smooth curve. The constants quoted above correspond to a potential with a maximum 1193 cm^{-1} above the minimum which occurs at an angle of 139°. However, it should be emphasized that observation of the lowest levels of the

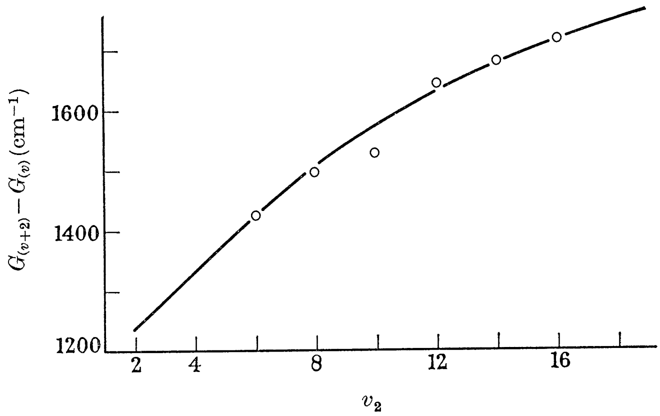

FIGURE 5. Intervals between successive Σ vibronic sublevels in the $\tilde{b}\,^1B_1$ state of CH_2 after deperturbation ($W_{ni} = 73$ cm^{-1} being used for the Fermi interaction constant). The curve represents the calculation described in the text.

excited state are needed to determine the constants more precisely and almost equally good fits can be obtained if C^2 is varied over the range 10 to 30 corresponding to maxima at 150 to 2500 cm^{-1} and angles at 131 to 154°. In any event, all the observed vibrational levels of the $\tilde{b}\,^1B_1$ state of CH_2 lie above the potential maximum, that is, for most practical purposes the observed levels of the molecule may be considered as those of a linear molecule.

When all the bands discussed so far had been assigned, a rather large number of observed lines remained unidentified. A number of the most intense of these unidentified lines were found to have nearly constant separations from lines of the main bands and could be quantitatively accounted for as belonging to hot bands with a lower level 1352·6 cm^{-1} above 0, 0, 0. The most natural explanation of this level is 0, 1, 0, that is, that $\nu_2'' = 1352{\cdot}6$ cm^{-1}. In agreement with this interpretation, the symmetry properties of the rotational levels of 0, 1, 0 are found to be the same as for the main bands. The presence of an appreciable fraction of CH_2 molecules in the 0, 1, 0 state is easily understood (even though the Boltzmann factor is much too small), since very probably the primary photodecomposition of diazomethane leads to highly excited CH_2 molecules which, only because of the presence of a great excess of inert gas, are quickly brought down to the lower vibrational levels. But for the first few vibrational levels the collisional transfer to the lowest level is generally very slow, so that a good fraction in the 0, 1, 0 level may still be left at the instant of observation. It may be noted that infrared transitions are even slower in removing vibrational energy since the transition probabilities are only of the order of 100 per second.

6. Determination of rotational constants

(a) *Lower state*, $\tilde{a}\,^1A_1$

In the $\tilde{a}\,^1A_1$ state the molecule is a strongly asymmetric top. For such a top (which is assumed to be rigid), the following simple expressions hold for the $J = 1$ and three of the $J = 2$ levels:

$$\left.\begin{aligned} F(1_{01}) &= B+C, \quad F(1_{11}) = A+C, \quad F(1_{10}) = A+B,\\ F(2_{12}) &= A+B+4C, \quad F(2_{11}) = A+4B+C, \quad F(2_{21}) = 4A+B+C.\end{aligned}\right\} \tag{10}$$

From these relations it follows immediately that

$$F(2_{21})-F(1_{01}) = 4A, \quad F(2_{11})-F(1_{11}) = 4B, \quad F(2_{12})-F(1_{10}) = 4C. \tag{11}$$

These differences can be obtained directly from the spectrum as differences of fairly strong lines in the subbands with $K' = 1$, $K' = 0$ and $K' = 2$ respectively as follows:

$$\left.\begin{aligned} {}^rR(1_{01})-{}^rQ(2_{21}) &= {}^rQ(1_{01})-{}^rP(2_{21}) = F(2_{21})-F(1_{01}),\\ {}^rQ(1_{11})-{}^rP(2_{11}) &= {}^rR(1_{11})-{}^rQ(2_{11}) = F(2_{11})-F(1_{11}),\\ {}^rR(1_{10})-{}^rQ(2_{12}) &= {}^rR(1_{10})-{}^rQ(2_{12}) = F(2_{12})-F(1_{10}).\end{aligned}\right\} \tag{12}$$

If the analysis is correct, these differences must come out to be exactly the same for different bands with the same lower state and give directly good approximations to $4A''$, $4B''$ and $4C''$ respectively. The only correction that has to be applied is the centrifugal distortion correction, but for these low rotational levels it is expected to to small, smaller indeed than the error involved in the determination of the differences. The resulting preliminary values of A''_0, B''_0, C''_0 were found to be 20·17, 11·20 and 7·07 respectively. Using these values, we obtain from equation (10) the energies of the three levels with $J = 1$ above $J = 0$.

By adding the combination differences (12) to the calculated $J = 1$ energies, and from the direct observation of the difference $F(2_{20})-F(0_{00})$, all but one of the group of $J = 2$ levels were evaluated directly from the spectrum. In the same way, by adding successively the combination differences between the $J = 3$ and $J = 2$, the $J = 4$ and $J = 3$, etc., levels, the energies of most of the rotational levels of the $\tilde{a}\,^1A_1$ state, which give rise to observed lines, can be established. The resulting values must be considered as 'observed' energy levels since they are based on observed line pairs, and since the only underlying assumption is that the $J = 1$ levels are given by the rigid asymmetric top formulae; an assumption that would be invalid only if the $J = 1$ levels were perturbed by another electronic state.

We have used the preliminary A, B, C values to calculate a set of energy levels by means of standard procedures for the rigid asymmetric top. The observed energy levels show systematic deviations from these calculated values. The deviations increase rapidly with K_a and are clearly due to the effect of centrifugal distortion.

The energy levels of a non-rigid rotator have been discussed by Wilson (1937) and Kivelson & Wilson (1952, 1953). A convenient form of the matrix elements may be found in Allen & Olson (1962). They are expressed in terms of the six constants D_J, D_{JK}, D_K, D_1, D_2, D_3, of which the last three (previously called δ_J, R_5, R_6 respectively) vanish for a symmetric top. For a planar asymmetric top molecule there are two relations between the six D values so that only four constants need be determined (see Kivelson & Wilson 1953; also Dowling (1961)). It is possible to determine these four constants as well as the rotational constants A, B, C by an application of Mecke's sum rules modified to take account of centrifugal stretching (see Allen & Olson 1962; Herzberg 1966). However, this method does not lead to the most precise values of the constants since only for the very lowest J values are all sublevels observed, and therefore only these can be used in the sum rules. Instead, we have used an iterative method involving the direct calculation of the energy levels by means of a computer program, and improving in each step the accuracy of the constants by means of Taylor's theorem.

Each rotational energy level, $F(J_{K_a K_c})$, is a function, $f(A, B, C, D_J, \ldots)$, of the rotational constants. According to Taylor's theorem, if we vary the constants by small amounts $\Delta A, \Delta B, \Delta C, \Delta D_J \ldots$ the function f changes by

$$\Delta f = \Delta A \frac{\partial f}{\partial A} + \Delta B \frac{\partial f}{\partial B} + \Delta C \frac{\partial f}{\partial c} + \Delta D_J \frac{\partial f}{\partial D_J} + \ldots.$$

We can thus express the deviations of the observed energy levels from those calculated from the preliminary A, B, C values in terms of a set of linear equations of condition which can be solved for $\Delta A, \Delta B, \Delta C, \Delta D_J, \ldots$ by the method of least squares. In carrying out this program, it was found that the data were not sufficiently accurate to determine meaningful values for all six D constants. Only for two of them, D_K and D_{JK}, were values obtained which were substantially larger than the standard deviation. It was therefore decided to assume that

$$D_J = D_1 = D_2 = D_3 = 0.$$

In table 6 the final constants for the lowest vibrational level, 0, 0, 0, of the $\tilde{a}\,^1A_1$ state are presented. The A_0, B_0, C_0 values differ only slightly from the provisional values given four years ago (Herzberg 1961), but no centrifugal distortion constants were available at that time. In table 7 the energy levels calculated from the final constants are compared with the observed energy levels. It is seen that for most of

Table 6. Rotational constants (in cm^{-1}) of the lowest singlet state, $\tilde{a}\,^1A_1$, of CH_2

0, 0, 0, level

$A = 20{\cdot}144 \pm 0{\cdot}019$	$D_K = 0{\cdot}0143 \pm 0{\cdot}0012$
$B = 11{\cdot}162 \pm 0{\cdot}010$	$D_{JK} = -0{\cdot}0014 \pm 0{\cdot}0008$
$C = 7{\cdot}060 \pm 0{\cdot}005$	

0, 1, 0 level $\nu_0 = 1352{\cdot}64 \pm 0{\cdot}08$

$A = 21{\cdot}807$	$\alpha_2^A = -1{\cdot}663 \pm 0{\cdot}016$
$B = 11{\cdot}294$	$\alpha_2^B = -0{\cdot}132 \pm 0{\cdot}009$
$C = 6{\cdot}927$	$\alpha_2^C = +0{\cdot}133 \pm 0{\cdot}008$

the levels the agreement is within the accuracy of the determination of the observed levels, but there do remain a few levels for which this is not true; the reason for these discrepancies is not understood.

TABLE 7. OBSERVED AND CALCULATED ENERGY LEVELS OF THE LOWEST SINGLET STATE OF CH_2

level	obs.	calc.	level	obs.	calc.
1_{01}	(18·25)[a]	18·22	5_{05}	247·34	247·28
1_{11}	(27·18)	27·19	5_{15}	247·95	247·89
1_{10}	(31·31)	31·30	5_{14}	304·43	304·53
2_{02}	(53·61)	53·54	5_{24}	313·78	313·98
2_{12}	59·57	59·54	5_{23}	339·67	339·76
2_{11}	71·96	71·85	5_{33}	—	376·46
2_{21}	98·51	98·61	5_{32}	381·65	381·35
2_{20}	99·67	99·73	5_{42}	449·55	450·31
			5_{41}	449·86	450·58
3_{03}	104·31	104·19	5_{51}	542·61	542·44
3_{13}	107·52	107·43	5_{50}	542·61	542·44
3_{12}	131·99	131·84			
3_{22}	153·27	153·30	6_{16}	339·70	339·79
3_{21}	158·51	158·44	6_{15}	412·94	413·48
3_{31}	208·60	208·26	6_{25}	418·83	418·52
3_{30}	208·82	208·46	6_{34}	488·12	488·10
			6_{43}	563·90	563·43
4_{04}	(168·93)	168·87	6_{52}	655·24	655·19
4_{14}	170·46	170·35	7_{16}[b]	532·55	535·90
4_{13}	210·12	210·02	7_{25}	596·99	598·47
4_{23}	225·37	225·33	7_{34}	642·00	641·84
4_{22}	238·68	238·65	7_{52}	786·15	787·59
4_{32}	283·50	283·02			
4_{31}	284·49	284·37			
4_{41}	356·16	356·46			
4_{40}	356·37	356·49			

[a] Values in parentheses have been assumed.
[b] Experimental values for levels with $J = 7$ seem somewhat doubtful and were not included in the least squares analysis.

The rotational constants of the 0, 1, 0 level were evaluated in a similar way except that the centrifugal distortion constants were assumed to be the same as for the 0, 0, 0 level. The resulting A, B, C values are included in table 6. These values imply the α_2 values given in the last part of the table.

(b) *Upper state*

In the upper $\tilde{b}\,{}^1B_1$ state of the red bands, as we have seen in § 4, the molecule is only slightly bent and indeed all observed states are well above the potential maximum, corresponding to the linear conformation. We can therefore evaluate the rotational constant in the same way as for a strictly linear molecule.

We use as the rotational energy formula

$$F'(J) = B'J(J+1) - D'J^2(J+1)^2. \qquad (13)$$

The term $-B'l^2$ which would occur for $l \neq 0$ is combined with the vibrational term $g_{22}\,l^2$.

In order to determine B_v and D_v, we have not used the usual method of combination differences, since in many of the subbands only very few such differences can be formed. Instead, we have added the wavenumbers of the lines to the appropriate observed energy levels of the ground state (see table 7), thus obtaining directly the term values of the upper state. If these are plotted against $J(J+1)$, the intercepts with the ordinate axis give the $\nu_0^{\text{sub.}}$ values, the initial slopes the B_v values, and the deviations from straight lines the D_v values. The values so obtained are included in table 5. Since in many instances only very few levels of a given v_2, l have been observed, the derived constants are not as precise as one might wish.

As in other linear molecules, when the bending vibration is excited, the interaction of vibration and rotation produces two characteristic effects on the rotational levels:

(1) l type doubling for all rotational series with $l \neq 0$ and (2) l type resonance between rotational series differing by 2 in their l values. As expected, l type doubling is pronounced in all Π vibrational levels; it can be represented by

$$\Delta\nu = qJ(J+1). \tag{15}$$

The values of q are included in table 5. In figure 6 the q values are plotted as a function of v_2 and compared with the theoretical variation

$$q = \frac{B^2}{\omega_2}\left\{1+\frac{4\omega_2^2}{\omega_3^2-\omega_2^2}\right\}(v_2+1) \tag{16}$$

in which we have set $B = 7{\cdot}7$, $\omega_2 = 810$ and $\omega_3 \simeq 3000$. There is a considerable downward deviation for low v_2, so much so that $q(v_2 = 9)$ has the opposite sign to that of the q values at higher v. These deviations must be ascribed to perturbation

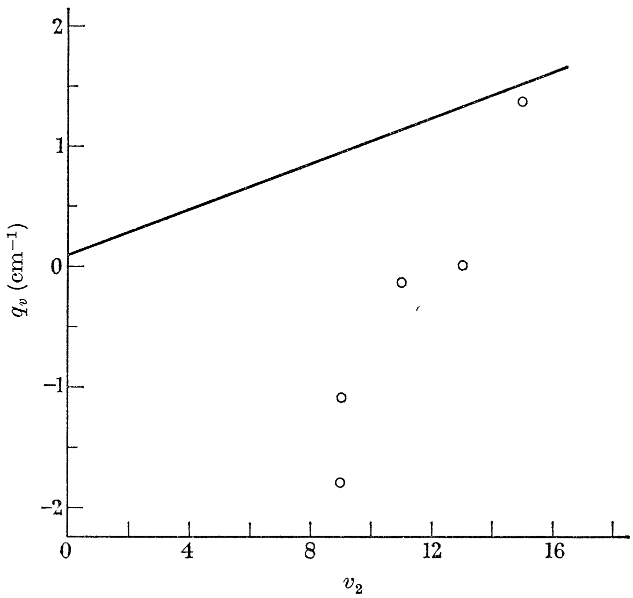

FIGURE 6. The l type doubling constant q as a function of v_2 in the $\tilde{b}\,^1B_1$ state of CH_2. The solid line represents the theoretical expectation.

by the $K = 1$ levels of the ground state for which the doubling has the opposite sign to what would be normal for the $\tilde{b}\,^1B_1$ state.

The vibrational levels with $l = 2$ also show a doubling which is, however, much smaller than for $l = 1$ and increases with $J^2(J+1)^2$. This splitting would be negligibly small if it were not for the fact that each Δ substate is accompanied by a Σ substate which interacts with one component of the Δ state, a phenomenon designated l type resonance by Nielsen (1959). The most striking effect of this phenomenon is observed for the corresponding Σ substates which since they lie above Δ, are pushed up by an amount that is proportional to $J^2(J+1)^2$. This effect is larger and opposite in sign to the effect of centrifugal distortion, and causes the apparent D values for all the Σ levels to be negative (see table 5).

7. Geometrical and electronic structure

The geometrical structure of CH_2 in the *equilibrium* positions of the $\tilde{a}\,^1A_1$ and $\tilde{b}\,^1B_1$ states cannot be determined since not all the α values have been determined, and thus the equilibrium rotational constants are not known. For the lower state for which A_0, B_0, C_0 have been determined, we can use either A_0, B_0 or A_0, C_0 or B_0, C_0 to determine the two geometrical parameters r_0 and θ_0. Because of the presence of an appreciable inertial defect, the three pairs of values will differ somewhat. Since, in the case of H_2O which is geometrically very similar to CH_2, the combination A_0, B_0 leads to r_0, θ_0 values closest to the equilibrium values, we choose A_0 and B_0 for the determination of the geometrical parameters in the $\tilde{a}\,^1A_1$ state of CH_2 and find

$$\tilde{a}\,^1A_1 \quad r_0 = 1{\cdot}11_1, \quad \theta_0 = 102{\cdot}_4{}^\circ.$$

In the upper state $\tilde{b}\,^1B_1$ of the red bands, only B_0 is known (and it is an extrapolated value). If we combine it with the value of the angle derived from the vibrational structure (see § 5) we have

$$\tilde{b}\,^1B_1 \quad \text{or} \quad r_0(\mathrm{CH}) = 1{\cdot}05_3, \quad \theta_0 = 140 \pm 15^\circ.$$

Thus the change of r_0 in the $\tilde{b}$–$\tilde{a}$ transition is small and, in accordance with the Franck–Condon principle, the main progression is a progression in v_2', corresponding to the large change in angle.

In the upper state $\tilde{c}\,^1A_1$ of the near ultraviolet bands our observations are not sufficiently extensive to determine the geometry.

In figure 7 the correlation of electronic states between linear and bent CH_2 is shown. The only low lying states of linear CH_2 are $^3\Sigma_g^-$, $^1\Delta_g$ and $^1\Sigma_g^+$, resulting from the electron configuration

$$1\sigma_g^2\, 2\sigma_g^2\, 1\sigma_u^2\, 1\pi_u^2.$$

The corresponding low lying states of bent CH_2 are $^1A_1$, $^3B_1$, $^1B_1$, $^1A_1$, resulting from the configurations

$$(1a_1)^2\,(2a_1)^2\,(1b_2)^2\,(3a_1)^2$$
$$(1a_1)^2\,(2a_1)^2\,(1b_2)^2\,(3a_1)\,(1b_1)$$
$$(1a_1)^2\,(2a_1)^2\,(1b_2)^2\,(1b_1)^2.$$

As shown by figure 7 the $^3\Sigma_g^-$ state of linear CH_2, which forms the ground state of the molecule, correlates with $^3B_1$ of bent CH_2; the $^1\Delta$ state correlates with the lower $^1A_1$ state and the $^1B_1$ state, and these states must be the lower and upper states of the red bands. The splitting of the $^1\Delta_g$ state on bending may be thought of as produced by a strong Renner–Teller (vibronic) interaction in the linear conformation similar to that previously discussed by Dressler & Ramsay (1959), and Pople & Longuet-Higgins (1958) for NH_2 in the $^2\Pi$ ground state of the linear conformation. Figure 8 shows the potential energy of the two states of CH_2 as a function of the bending angle.

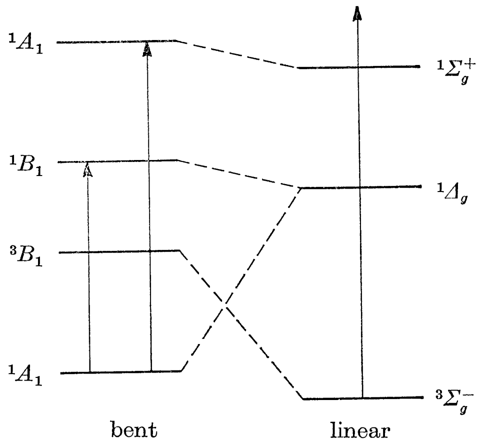

FIGURE 7. Correlation of electronic states of linear and bent CH_2. Observed transitions have been indicated by vertical arrows.

The upper state of the near ultraviolet bands must be the $^1A_1$ state that correlates with $^1\Sigma_g^+$ of the linear conformation. Indeed, since in all probability the observed state does not have a hump in the linear conformation, it is better described as the $^1\Sigma_g^+$ state.

Thirteen years ago Walsh (1953) made some predictions about the geometrical structure of XH_2 molecules in their low lying electronic states on the basis of diagrams (Walsh diagrams) giving the variation of the orbital energies with angle. On this basis the bent structure of H_2O is ascribed to the two electrons in the $3a_1$ orbital whose energy decreases rapidly when the molecule is bent. The two electrons in the outermost orbital $1b_1$ do not change their energy much when the molecule is bent; they are essentially the 'lone-pair' electrons of the O atom. Both in the ground state of NH_2 and the lowest singlet state of CH_2 there are still two electrons in the $3a_1$ orbital, and therefore Walsh predicted these molecules to have an angle of bend similar to that in the ground state of H_2O. This prediction is strikingly borne out by the observations on NH_2 and CH_2. In the first excited state of NH_2 and the first excited singlet state of CH_2, one electron is removed from the $3a_1$ orbital and brought to the $1b_1$ orbital. Therefore Walsh predicted these states to be much less strongly bent as indeed they are observed to be. In fact, all the observed vibrational levels are above the potential hump, corresponding to the linear conformation, that is, NH_2 and CH_2 in these states are quasi-linear.

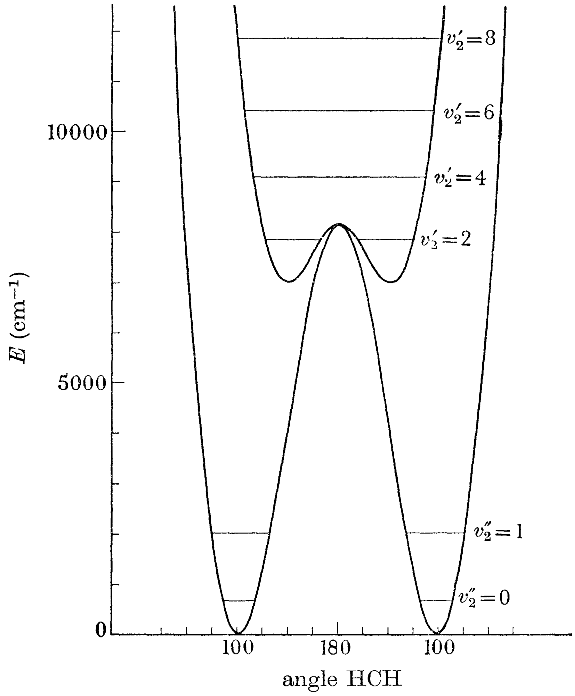

FIGURE 8. Potential energy curves of the two lowest singlet states of CH_2. The vibrational quantum number, v_2', of the upper state has been assigned assuming this state to be linear.

8. CONCLUSION

While the preceding study has firmly established the nature of the lowest and of the first excited singlet states of CH_2, it has not been possible to obtain much information about other singlet states nor has it been possible to establish the exact relation of the singlet states to the triplet states. For the latter purpose it would be necessary either to observe an intercombination system between a triplet and a singlet state or to identify perturbations produced in one of the singlet states by one of the triplet states. For either purpose absorption spectra of much greater intensity will be required. The only method to accomplish this seems to be that of substantially increasing the length of the absorbing path. However, there can be very little doubt, on the basis of the argument previously presented (Herzberg 1961), that the lowest triplet state ($\tilde{X}\,^3\Sigma_g^-$) lies below the lowest singlet state ($\tilde{a}\,^1A_1$). In addition to the spectroscopic arguments, there is now a good deal of information (see, for example, Frey (1960)) about the reactions of singlet and triplet CH_2 which confirm the conclusion that triplet CH_2 (in its lowest state) has a substantially longer life than singlet CH_2 (in its lowest state) or, in other words, that $\tilde{X}\,^3\Sigma_g^-$ is the ground state of CH_2.

The features observed in the red bands of CH_2 agree in almost every respect with the theoretical prediction that the two lowest singlet states of CH_2 are derived from a single $^1\Delta_g$ state in the linear conformation which is split by a large Renner–Teller interaction. While both components have a bent equilibrium conformation the potential hump in the upper component is equivalent to only about one vibrational quantum.

The near ultraviolet bands of CH_2 were found to be too weak for any analysis to be made but it seems very probable that the upper state of these bands is the predicted ${}^1\Sigma_g^+$ state. The absorption features in the vacuum ultraviolet near 1400 Å which behave like the other singlet features are too diffuse for analysis.

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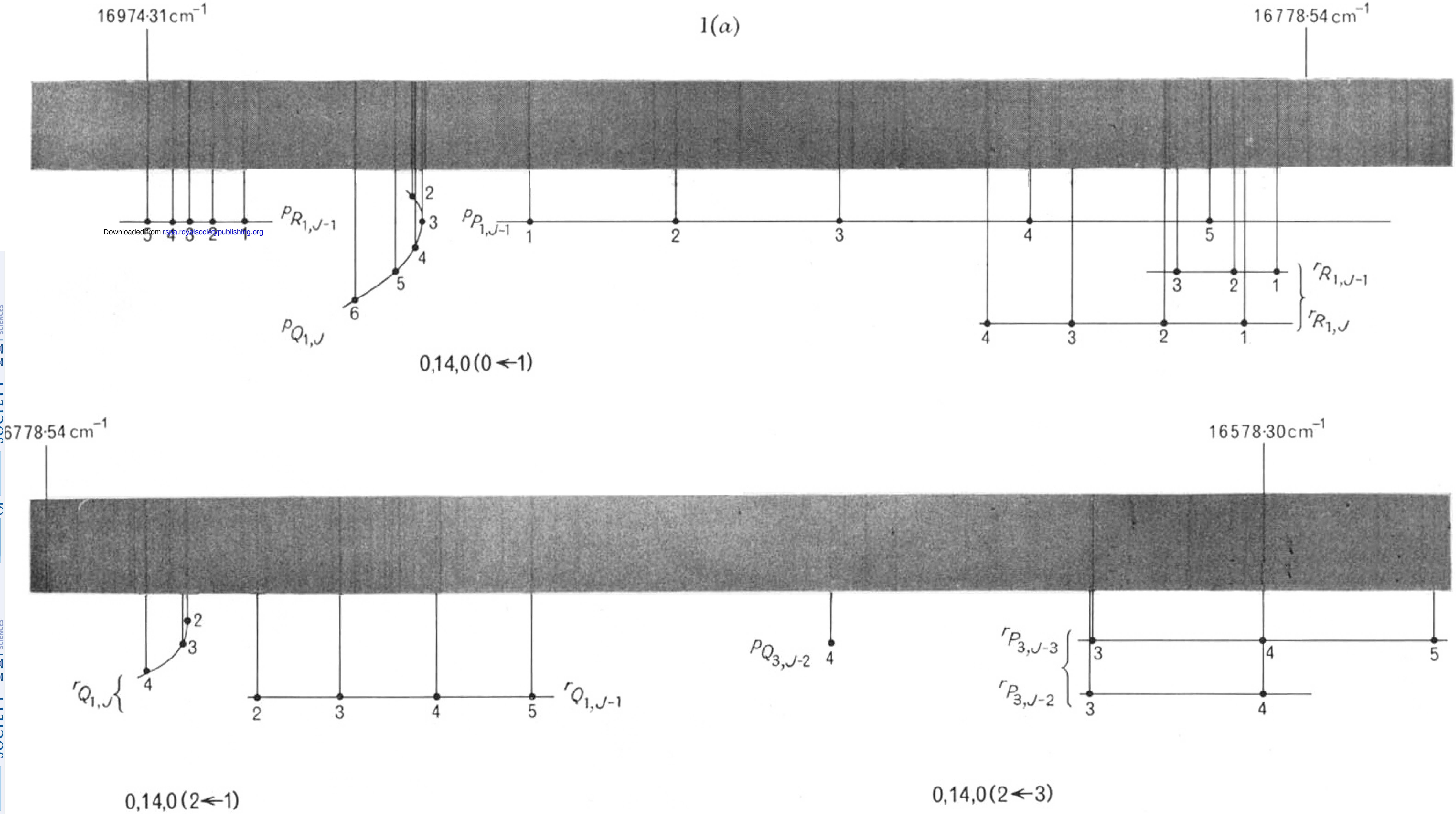

FIGURE 1 (*a*). Absorption band of CH_2 near 5900 Å. Assignments to the Σ–Π, Δ–Π and Δ–Φ subbands are indicated.

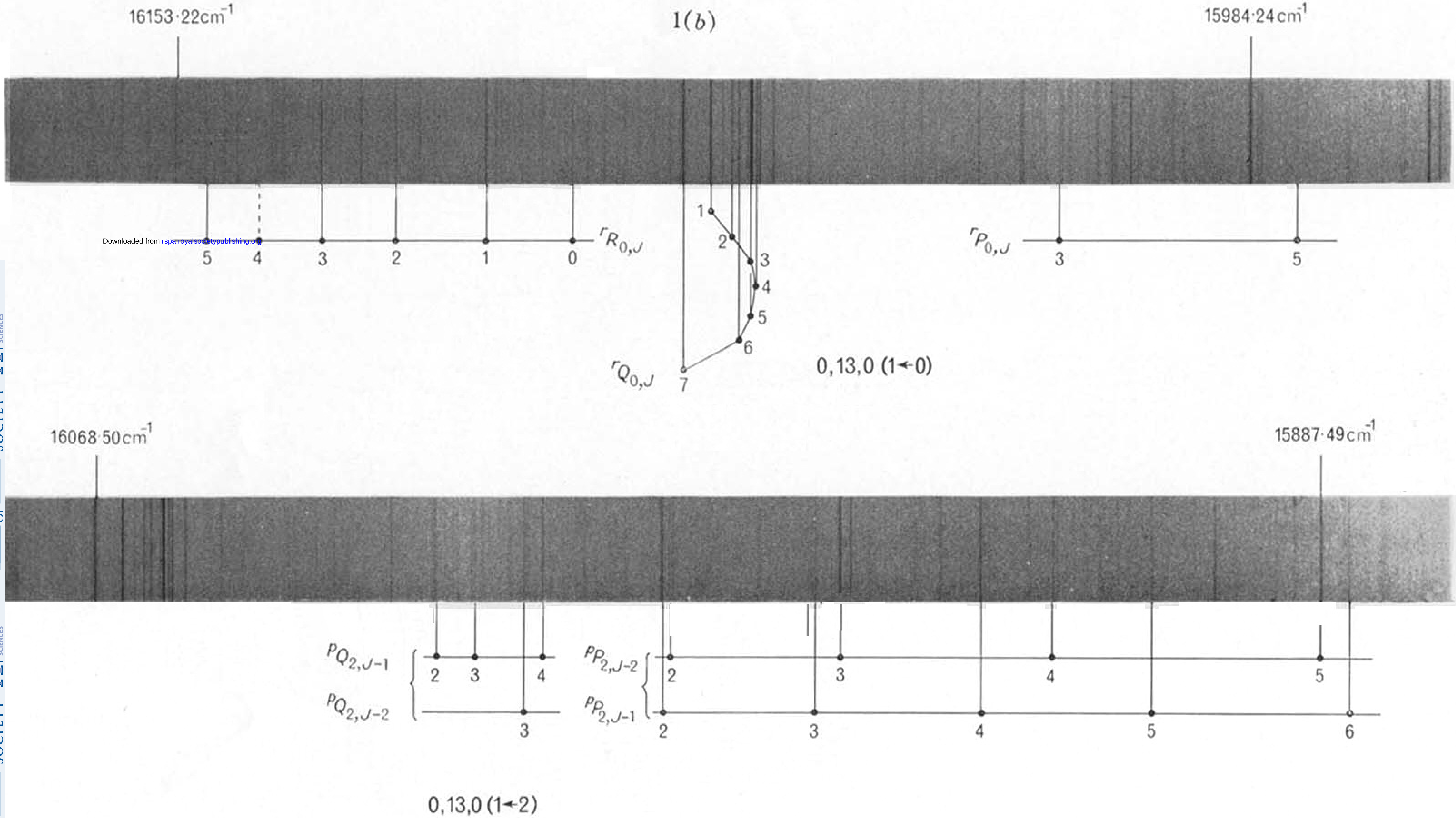

FIGURE 1(*b*). Absorption band of CH_2 near 6220 Å. Assignments to the Π–Σ and Π–Δ subbands are indicated.

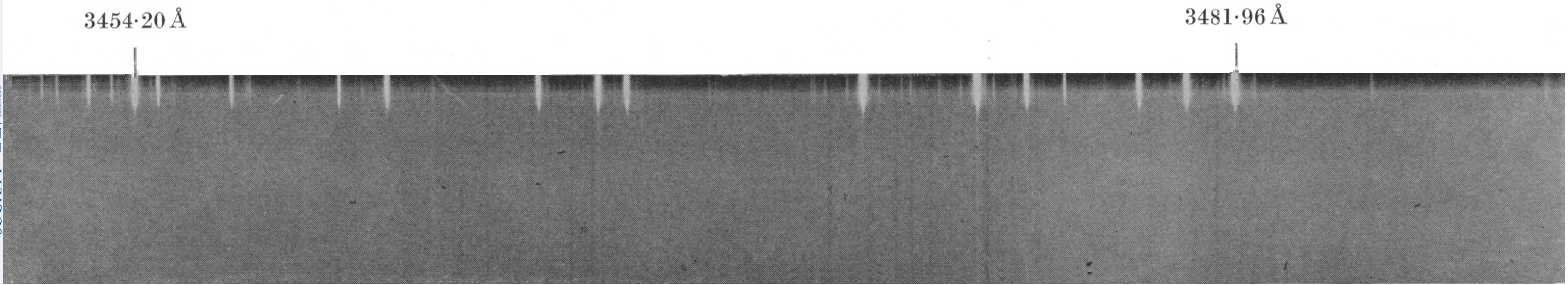

FIGURE 2. Absorption band of CH_2 near 3470 Å. This is the strongest of the near ultraviolet bands.